Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/04/2007
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Project Title:	Checkpoint Reduced Cell Sensitivity to High Energy Particles-Induced Killing		
Division Name:	Human Research		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 4/2009) NOTE: Received NCE to 6/30/2008 per A. Chu (6/07) NOTE: End date changed to 6/30/07 per A. Chu (5/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	This application is in response to the NASA Program NRA 02-OBPR-02 and addresses research related to both Bioastronautics and Fundamental Biology. Radiation-induced DNA double strand breaks (DSBs) are a severe threat to genomic integrity that determines radiosensitivity (short term effect) and carcinogenesis (long term effect). Two major DNA DSB repair pathways exist in eukaryotic cells, homologous recombination repair (HRR) and non-homologous end joining (NHEJ). After induction of DSBs, proliferating cells actively slow down cell cycle progression via checkpoint activation to provide time for repair. However, it is not known whether checkpoint activation benefits NHEJ and HRR equally, or whether it is specifically adapted to assist just one of these processes, therefore reducing DSBs-induced genomic instability. The long-term objective of the present proposal is to elucidate the molecular mechanisms by which checkpoint activation facilitates DNA repair and therefore reduces the high-energy (HZE) particles-induced genomic integrity in mammalian cells. In this proposal, the central hypothesis is that CHK1-regulated checkpoints, in an ATR-dependent but ATM-independent manner, facilitate HRR but not NHEJ, thus affecting HZE particles-induced killing of mammalian cells. The preliminary results from sparsely ionizing radiation provide evidence to support this hypothesis and form the foundation for the experimental design. Three Specific Aims are pursued: Specific aim 1: Determine whether ATR, the upstream kinase of CHK1, affects HZE particles-induced killing by facilitating HRR. We will examine HRR (by using the GFP-I-SceI restriction enzyme system), NHEJ (by using asymmetric field inversion gel electrophoresis, AFIGE), CHK1 activity (by using kinase assays) and the effect of CHK1 siRNA on HZE particles-induced killing (by using clonogenic assay) in ATR kinase dead (ATR-kd) human fibroblasts. Specific aim 2: Determine whether the protective effect of the CHK1-regulated checkpoint on HZE particles-induced killing is mediated by facilitating HRR. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to A1-5 cells, a rat embryo fibroblast line in which CHK1 is highly expressed. We will examine HRR, NHEJ, CHK1 activity and HZE particles-induced killing in CHK1-kd cell lines. Specific aim 3: Determine whether the effect of CHK1 on HZE particles-induced killing is ATM independent. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to AT5BIVA (ATM-/-) human fibroblast cells. We will test HZE particles-induced killing, HRR, NHEJ, CHK1 activity and checkpoint activation in the CHK1-kd AT cell lines following HZE particles radiation. This approach will clarify the relationships among ATR/CHK1-regulated checkpoint activation, NHEJ and HRR. Such information will provide means for protecting mammals from HZE particles-induced killing in a short term and from HZE particles-induced carcinogenesis in a long term by activating a particular pathway (ATM or ATR-regulated checkpoints, NHEJ or HRR) depending on the genetic background of the mammals. The proposal addresses a basic research topic related to the protection of individuals from the hazards of ionizing radiation in outer space. The proposed basic research is within the framework defined by the Space Radiation Health Program, and attempts to establish the scientific basis for radiation protection of humans engaged in the exploration of outer space. It is a ground-based program that studies the hypothesis that CHK1-regulated checkpoints, in an ATR-dependent but ATM-independent manner, facilitate HRR but not NHEJ, thus affecting HZE particles-induced killing of mammalian cells. We believe that our results from this proposal will clarify the relationships among ATR/CHK1-regulated checkpoint activation, NHEJ and HRR. Once we understand the relationships among checkpoint activation, HRR and NHEJ, we will be able to protect cells from HZE particles-induced killing by activating the particular pathway (ATM- or ATR/CHK-regulated checkpoint, HRR or NHEJ) of the cells with different genetic backgrounds. We expect that these results will have the potential for reducing the HZE particles-induced mutagenesis and carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results from these experiments will clarify the role of the ATR/CHK1 pathway, the most important checkpoint proteins in mammalian cells, in protecting cells from ionizing radiation (IR)-induced killing.
Task Progress:	This project has been funded for four years (02-OBPR-02-0000-0037). We have followed the proposal to carry out some experiments as described in our specific aims. Radiation-induced DNA double strand breaks (DSBs) are a severe threat to genomic integrity that determines radiosensitivity (short-term effect) and carcinogenesis (long-term effect). Two major DNA DSB repair pathways exist in eukaryotic cells, homologous recombination repair (HRR) and non-homologous end joining (NHEJ). After induction of DSBs, proliferating cells actively slow down cell cycle progression via checkpoint activation to provide time for repair. However, it is not known whether checkpoint activation benefits NHEJ and HRR equally, or whether it is specifically adapted to assist just one of these processes therefore reducing DSBs-induced genomic instability. The long-term objective of the present proposal is to elucidate the molecular mechanisms by which checkpoint activation facilitates DNA repair and therefore reduces the high-energy (HZE) particles-induced genomic integrity in mammalian cells. In this proposal, the central hypothesis is that CHK1-regulated checkpoints, in an ATR-dependent but ATM-independent manner, facilitate HRR but not NHEJ, thus affecting HZE particles-induced killing of mammalian cells. The preliminary results from sparsely ionizing radiation provide evidence to support this hypothesis and form the foundation for the experimental design. Three Specific Aims are pursued: Specific aim 1: Determine whether ATR, the upstream kinase of CHK1, affects HZE particles-induced killing by facilitating HRR. We will examine HRR (by using the GFP-I-SceI restriction enzyme system), NHEJ (by using asymmetric field inversion gel electrophoresis, AFIGE), CHK1 phosphorylation (by using Western blot) and the effect of CHK1 siRNA on HZE particles-induced killing (by using clonogenic assay) in ATR kinase dead (ATR-kd) human fibroblasts. Specific aim 2: Determine whether the protective effect of the CHK1-regulated checkpoint on HZE particles-induced killing is mediated by facilitating HRR. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to GM847, the parental cell line of ATR-kd. We will examine HRR, NHEJ, CHK1 activity and HZE particles-induced killing in CHK1-kd cell lines. Specific aim 3: Determine whether the effect of CHK1 on HZE particles-induced killing is ATM independent. We will transfect the CHK1-kd constructs combined with Tet-OnTM systems to AT5BIVA (ATM-/-) human fibroblast cells. We will test HZE particles-induced killing, HRR, NHEJ, CHK1 phosphorylation and checkpoint activation in the CHK1-kd AT cell lines following HZE particles radiation. This approach will clarify the relationships among ATR/CHK1-regulated checkpoint activation, NHEJ and HRR. Such information will provide the means for protecting mammals from HZE particles-induced killing in a short-term and from HZE particles-induced carcinogenesis in a long-term by activating a particular pathway (ATM or ATR-regulated checkpoints, NHEJ or HRR) depending on the genetic background of the

	mammals. During the past four years we have finished specific aims 1 and 2, and partial specific aim 3. In the fourth year of the proposal (7/1/2006-6/30/2007), we successfully performed the experiments in Brookhaven National Laboratory by using the NASA beam facility located there. During these experiments, we focused on the third specific aim that determines whether the effect of CHK1 on HZE particles-induced killing is ATM independent. By combining with some data obtained from X-ray irradiated cells, we obtained some results that provided strong evidences to support our hypothesis. However, we still need more experiments to finish our goals described in this grant. Therefore, we request no-cost extension for one more year about this grant. We previously showed that high LET IR inhibits only Ku-dependent NHEJ but inhibits neither PARP-1 dependent NHEJ nor HRR. These results provide an evaluation on whether one gene is involved in Ku-dependent NHEJ. If a gene is involved in Ku dependent NHEJ, high LET induced RBE on killing of the cell line deficient in the gene should be lower than the RBE on killing of its wild type counterpart. Based on this conclusion, we examined the survival sensitivity of ATM deficient cell line to either high LET IR or low LET IR. The results show that high LET does kill more ATM deficient cells than X-ray does, the RBE on killing in ATM deficient cells is similar to that in the wild type counterpart. These results suggest that ATM is not directly involved in Ku dependent NHEJ. Next step, we will examine the effects of DNA-PK siRNA or Artemis siRNA on the sensitivity of ATM deficient cells to IR. These results will provide direct evidence whether ATM is directly involved in Ku-dependent NHEJ.
Bibliography Type:	Description: (Last Updated: 07/07/2021)
Abstracts for Journals and Proceedings	Wang X, Wang Y. "MEPE/OF45 is a Substrate of CHK1 Kinase to Respond to DNA Damage." Presented at the 53rd Annual Meeting of the Radiation Research Society, Philadelphia, PA, November 5th–8th, 2006. 2006 Meeting Abstracts, 53rd Annual Meeting of the Radiation Research Society, 2006. , Nov-2006
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